

ASHRAE Modified Hunter Curve - Flow Charts

Chart 12-1

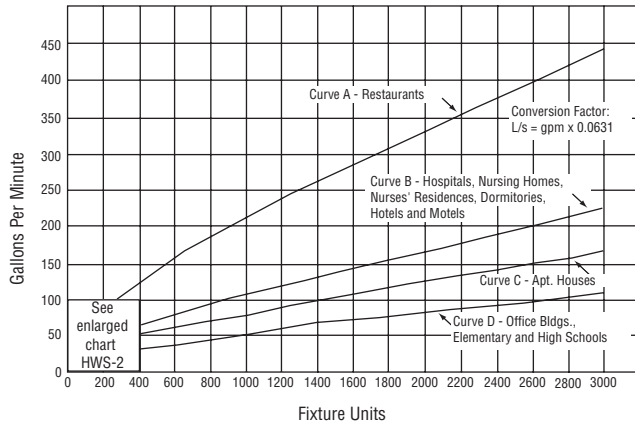
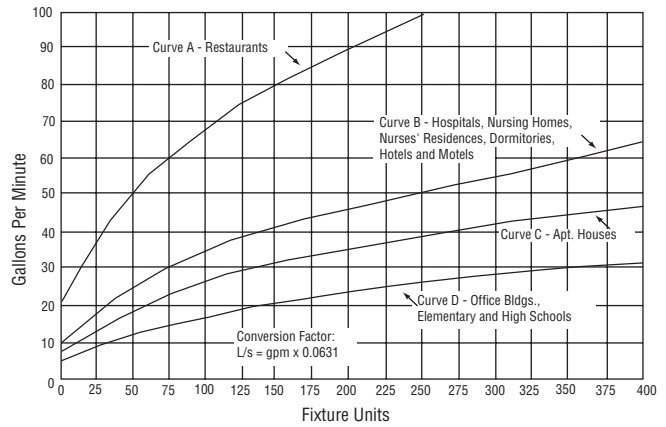


Chart 12-2. Enlarged Section



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Step 1

Determine the total fixture unit load for all the fixtures serviced by your water heater application using the Fixture Units Table on page 13. See example below.

Step 2

Using the total fixture units for your application, enter the Hunter Curves (Chart 12-1) from the bottom on the total fixture units line for your application. Read up to the curve that best fits the application. Then read to the left for the corresponding gpm requirement.

Step 3

Select the proper Armstrong Water Heater/Water Temperature Controller.

Example: College Dormitory

No. Fixtures	Type of Fixture	Fix. Unit	Demand Fix. Unit
150	Private Lavatory	.75	113
120	Private Shower	1.5	180
20	Slop Basin	2.5	50
8	Clothes Washer	2.0	16
Total Fixture Units			359

Refer to the modified Hunter Curves in Chart 12-2. Curve B represents dormitories. Enter the graph from the bottom at 359 fixture units and go up to curve B. Then move to the left horizontally to read approximately 60 gallons per minute of hot water capacity required.

NOTE: Remember to add any constant flow capacities, as determined under "Important Note" below, to this 60 gpm.

Important Note

Special consideration should be given to applications involving periodic use of gang showers, process equipment, laundry machines, etc., as may occur in field houses, gymnasiums, factories, hospitals, etc. Because these applications could have all equipment on at the same time, their total hot water capacity should be determined and then added to the maximum hot water demand as read from the modified Hunter Curves. Use the following formula to determine total hot water capacity needed for these applications when final water temperatures are lower than that of the water heater.

$$\frac{(B - C)}{(H - C)} \times \left(\text{Total water flow from all gang shower heads in gpm} \right) = \text{Hot water needed (gpm)}$$

Where:

B = Blended water temperature out of the fixture

H = Hot water temperature to the fixture

C = Cold water temperature to the fixture